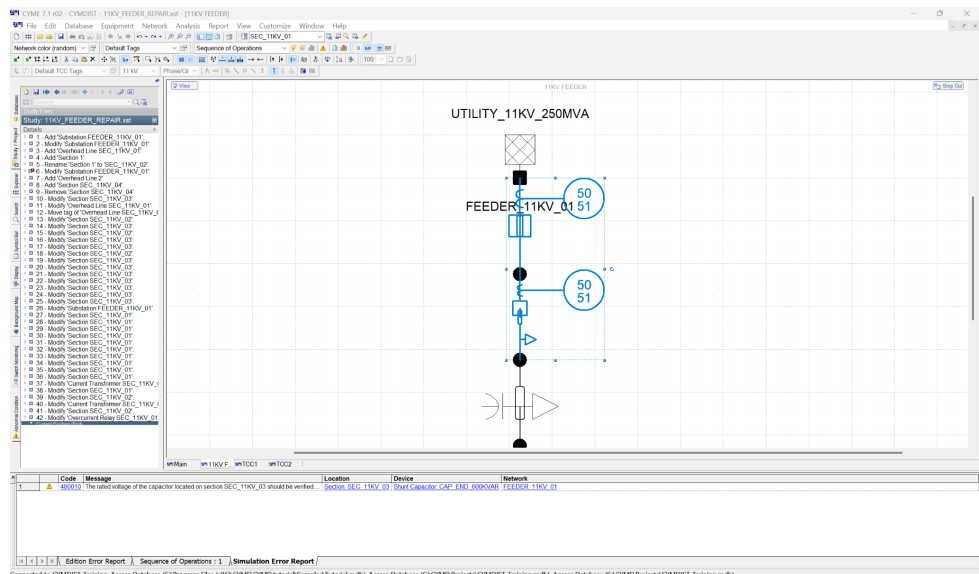


Protection Coordination Summary - 11 kV CYME Feeder

CYME/CYMDIST phase-overcurrent coordination using MULTILIN 350 IEC Short Inverse relays

System Overview



Final Protection Settings

Relay / breaker	CT ratio	Primary pickup	Time dial	Role
SEC_11KV_01	600/5 A	550 A	0.40	Upstream backup
SEC_11KV_02	600/5 A	550 A	0.05	Downstream primary

Coordination Verification

Test case	Fault current	Operating breaker	Trip time	Coordination result
Downstream LLL	4,574.7 A	SEC_11KV_02	0.078 s	Selective
Downstream LLG	4,231.4 A	SEC_11KV_02	0.079 s	Selective
Downstream LL	3,961.8 A	SEC_11KV_02	0.080 s	Selective
Downstream LG	3,060.6 A	SEC_11KV_02	0.085 s	Selective
Upstream LLL	8,741.2 A	SEC_11KV_01	0.221 s	Correct upstream operation

Reclosing

For the coordination study, the breaker sequence was configured for one operation to lockout. This removed repeated open-close cycles during persistent-fault simulations and allowed the protection selectivity to be assessed clearly.

Shunt Capacitor

The feeder model includes a fixed shunt capacitor bank at SEC_11KV_03. The final model shows a nominal rating of approximately **600 kVAr** at **11 kV**. The capacitor is included to represent reactive-power compensation and voltage support; it was not treated as a separate capacitor-optimization study.

Section Properties

Section ID

SEC_11KV_03

Phase

☒ A
☒ B
☒ C

Zone

UNDEFINED

Environment

Unknown

More...

Devices

Add

Remove

Nodes

Overhead Line Balanced

Spot Load Three Phase

Shunt Capacitor

Control

Collapse

Shunt Capacitor

Id:

CAP_600KVAR_11KV

Number:

CAP_END_600KVAR

Status:

Connected

Location:

At To Node

Stage:

Undefined

Settings

Type:

Fixed

Rated Voltage:

11.0

kVLN

Interrupting Rating:

600.0

A

Configuration

Fixed

Phase:

☒ A
☒ B
☒ C

Rated Power:

600.0

kvar

Total

Losses:

0.0

kW

Total

Switched

Phase:

☒ A
☒ B
☒ C

Rated Power:

600.0

kvar

Total

Losses:

0.0

kW

Total

OK

Cancel

11KV F...

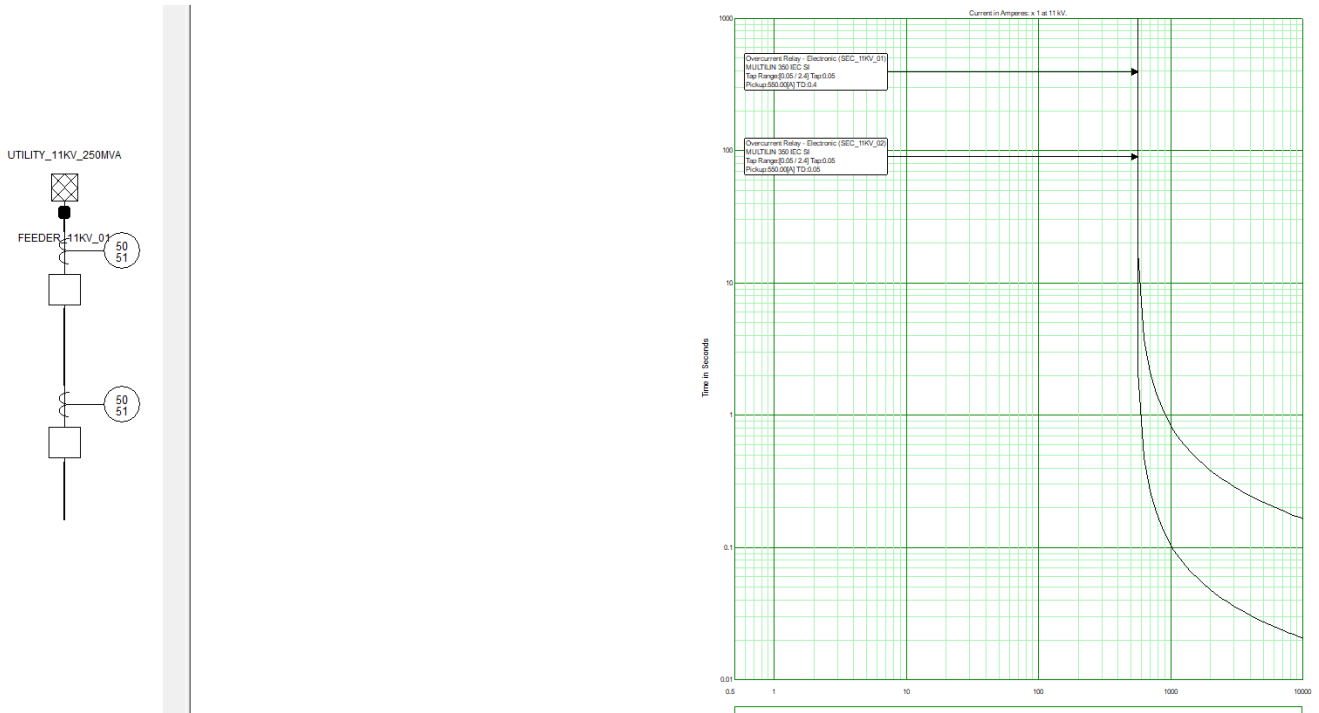
TCC1

TCC2

TCC Br...

TCC Br...

Combined TCC



The combined TCC demonstrates the intended grading: SEC_11KV_02 is the faster downstream device, while SEC_11KV_01 is delayed as upstream backup. The final verified time-dial settings were 0.05 downstream and 0.40 upstream.

Engineering Conclusion

The final CYME model achieved selective phase-overcurrent coordination for the tested LLL, LLG, LL, and LG downstream faults. A 600/5 A phase CT and 550 A primary pickup were used on both feeder relays. The downstream relay operated first for all tested downstream faults, while the upstream relay remained closed; the upstream breaker operated correctly for an upstream fault. The study also included a 600 kVAr shunt capacitor in the feeder model and resolved the breaker/reclosing configuration required for stable Sequence of Operations analysis.

Portfolio note: These settings belong to the modeled training system and are not intended for direct application to a real network without a complete engineering protection study and utility requirements review.